A

Project Report On

Nukkad

By

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of

B.Tech CE , Semester VI

Subject: System Design Practice

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CERTIFICATE

This is to certify that the practical/term work carried out in the subject of System Design Practice and recorded in this report is the bonafide work of

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Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to describe the requirements of the **Nukkad Business Directory application**. This document explains the features, functionality, and constraints of the system.

The Nukkad application is a **mobile-first platform** that helps customers find and connect with nearby local businesses. It also allows business owners to create business profiles, manage offers, and respond to customer enquiries.

This SRS document covers the **complete functionality of the application**, including user authentication, business discovery, enquiry management, subscription management, emergency directory access, and analytics features. It serves as a reference for developers, testers, and project stakeholders during development and evaluation.

1.2 Document Conventions

This document follows a structured format based on the **IEEE 830 Software Requirements Specification standard**.
The following conventions are used in this document:

- Section headings and numbering are used to organize the content clearly.
- Functional requirements are labeled using identifiers such as **FR-1, FR-2, etc.**
- System requirements are written using the phrase **"The system shall…"** to clearly define expected behavior.
- Diagrams such as **Use Case Diagrams, ER Diagrams, and Data Flow Diagrams** are included to visually represent system processes and data relationships.

These conventions help ensure that the document is easy to read and understand for both technical and non-technical readers.

1.3 Intended Audience and Reading Suggestions

This document is intended for the following audience:

- **Developers** – to understand the system features and implement the application.
- **Project reviewers or professors** – to evaluate the project requirements and design.
- **Testers** – to understand the expected system behavior and create test cases.
- **Business owners and users** – to understand how the system works and what services it provides.

The document is organized into several sections. The **Introduction** section provides an overview of the project and the purpose of the document. The **Overall Description** section explains the system environment, users, and general functionality. The **Specific Requirements** section describes detailed functional and non-functional requirements of the system. Finally, the **Appendices and Diagrams** provide information about the database structure and system architecture.
Readers are encouraged to begin with the introduction and then proceed to the overall description before reading the detailed requirements.

1.4 Project Scope

The **Nukkad Business Directory** is a digital platform designed to connect customers with nearby local businesses in a simple and efficient way.

Customers can search for businesses by category or location, view business profiles, explore available offers, and send enquiries to business owners. They can also save businesses for future reference and leave reviews based on their experience.

Business owners can register their businesses, update their profiles, create promotional offers, and respond to customer enquiries. The system also includes a **subscription model**, where users can either use limited free enquiry credits or subscribe to a yearly plan that provides unlimited connects.

In addition, the application provides an **emergency directory** that allows users to quickly access important emergency contact numbers such as police, ambulance, and fire services.

The goal of the system is to **improve visibility for local businesses and make it easier for customers to discover and contact nearby services**.

1.5 References

The following documents and resources were used as references while preparing this Software Requirements Specification:

- **Supabase Database Schema** – supabase_complete_schema.sql
- **Project Completion Summary Document** – PROJECT_COMPLETION_SUMMARY.md
- **IEEE 830 Software Requirements Specification Standard**
- **Flutter Documentation** – https://flutter.dev
- **Supabase Documentation** – https://supabase.com/docs

These resources provide information about the system architecture, database structure, and technologies used in the development of the application.

1.6 Impact Perspective

The *Nukkad Business Directory* improves business visibility by helping local businesses create online profiles and reach nearby customers easily. It enhances user engagement by allowing users to search, explore offers, and directly connect with businesses through enquiries, making the platform interactive and convenient to use.

2. Overall Description

2.1 Product Perspective

The **Nukkad Business Directory** is a new, standalone mobile application designed to help users discover and connect with local businesses easily.

The system consists of a **Flutter-based mobile application** connected to a **Supabase backend with a PostgreSQL database**. The application interacts with device services such as geolocation for finding nearby businesses and the phone dialer for contacting businesses or emergency services.

The main components of the system are:

- **Flutter Mobile Application** – User interface and application logic
- **Supabase Backend** – API and authentication services
- **PostgreSQL Database** – Stores users, businesses, enquiries, offers, and other data
- **Device Services** – Location services, phone dialer, and external links

2.2 Product Features

The major features of the Nukkad application include:

- **User authentication** for signup and login
- **Role selection** for customers and business owners
- **Business discovery** through search, filters, and categories
- **Business profile viewing** with details and offers
- **Customer enquiries** sent to businesses
- **Business profile and offer management** for business owners
- **Subscription and connect management**
- **Emergency numbers directory** for quick access to important contacts
- **Reviews and ratings** for businesses
- **Saved businesses** for future reference

2.3 User Classes and Characteristics

The system has three main user classes:

Customer

- Searches for businesses and services
- Sends enquiries and saves businesses
- Leaves reviews and views offers
- Uses emergency contact directory

Business Owner

- Creates and manages business profiles
- Posts and manages promotional offers
- Receives and responds to customer enquiries
- Tracks analytics and manages subscriptions

Admin

- Manages business categories and emergency numbers
- Moderates data and maintains system content

Customers and business owners are the **primary users** of the system.

2.4 Operating Environment

The software will operate in the following environment:

- **Platform:** Android, iOS, and Web (primarily Android)
- **Frontend:** Flutter mobile application
- **Backend:** Supabase backend services
- **Database:** PostgreSQL
- **Network:** Internet connection required for most features
- **Device Services:** GPS for location and phone dialer for calls

2.5 Design and Implementation Constraints

The system development has the following constraints:

- The frontend must be developed using **Flutter and Dart**.
- The backend and database must use **Supabase and PostgreSQL**.
- **Role-based access control** must be enforced using database policies.
- Internet connectivity is required for real-time features.
- Payment gateway integration is not fully implemented and transactions are stored using IDs.

2.6 User Documentation

The following documentation will be provided for users:

- **In-app onboarding screens** to guide new users
- **Help and usage instructions** within the application
- **Project documentation** including SRS and system diagrams

2.7 Assumptions and Dependencies

The system assumes the following:

- The **Supabase backend service is available and properly configured**.
- Users have **internet connectivity** while using the application.
- Device features such as **GPS and phone dialer** are available.
- Subscription payments will be handled externally and transaction IDs will be recorded in the system.

The project also depends on **Flutter, Supabase services, and PostgreSQL database infrastructure**.

3. System Features

3.1 User Authentication and Role Management

3.1.1 Description and Priority

This feature allows users to **sign up, log in, and select their role** as either a customer or a business owner. It ensures secure access to the system and personalized dashboards.

Priority: High

3.1.2 Stimulus / Response Sequences

- User opens the application.
- User selects **Sign Up** or **Login**.
- User enters phone number and password.
- System verifies the credentials.
- User selects a role (Customer or Business Owner).
- System stores user data and redirects to the appropriate dashboard.

3.1.3 Functional Requirements

REQ-1: The system shall allow users to register using phone number and password.
REQ-2: The system shall allow registered users to log in securely.
REQ-3: The system shall allow users to select a role (Customer or Business Owner).
REQ-4: The system shall store user profile information in the database.
REQ-5: The system shall display an error message if login credentials are incorrect.

3.2 Business Discovery and Viewing

3.2.1 Description and Priority

This feature allows customers to search, browse, and view local businesses by category or location.
Priority: High

3.2.2 Stimulus / Response Sequences

- Customer opens the search page.
- Customer enters keywords or selects a category.
- System retrieves matching businesses from the database.
- System displays a list of businesses.
- Customer selects a business to view its full profile.

3.2.3 Functional Requirements

REQ-1: The system shall allow customers to search businesses by name or category.
REQ-2: The system shall display a list of businesses matching the search query.
REQ-3: The system shall allow users to view detailed business profiles.
REQ-4: The system shall display business information including name, category, contact details, and offers.
REQ-5: The system shall display a message if no results are found.

3.3 Enquiry and Communication Management

3.3.1 Description and Priority

This feature allows customers to send enquiries to businesses and allows business owners to view and respond to enquiries.
Priority: High

3.3.2 Stimulus / Response Sequences

- Customer views a business profile.
- Customer fills out the enquiry form.
- Customer submits the enquiry.
- System stores the enquiry in the database.
- Business owner receives the enquiry and views it in their dashboard.
- Business owner responds to the enquiry.

3.3.3 Functional Requirements

REQ-1: The system shall allow customers to create enquiries for businesses.
REQ-2: The system shall store enquiries in the database.
REQ-3: The system shall deduct one connect when an enquiry is created.
REQ-4: The system shall allow business owners to view received enquiries.
REQ-5: The system shall allow business owners to respond to enquiries.

3.4 Business Profile and Offers Management

3.4.1 Description and Priority

This feature allows business owners to create and manage business profiles and promotional offers.
Priority: Medium

3.4.2 Stimulus / Response Sequences

- Business owner logs into the application.
- Business owner opens the business dashboard.
- Business owner creates or edits a business profile.
- Business owner adds or updates promotional offers.
- System saves the information in the database.

3.4.3 Functional Requirements

REQ-1: The system shall allow business owners to create a business profile.
REQ-2: The system shall allow business owners to update business details.
REQ-3: The system shall allow business owners to create promotional offers.
REQ-4: The system shall allow business owners to edit or deactivate offers.
REQ-5: The system shall store business and offer information in the database.

4. External Interface Requirements

4.1 User Interfaces

The application provides a **mobile user interface built using Flutter**. The UI is designed to be simple and mobile-friendly.
Main screens include:

- **Onboarding and Login/Signup screens** for user authentication
- **Customer Dashboard** for searching and viewing businesses
- **Business Owner Dashboard** for managing profiles, enquiries, and offers
- **Search and Filter screens** for discovering businesses
- **Business Profile page** showing business details, offers, and contact options
- **Enquiry Form** for customers to send enquiries
- **Emergency Numbers page** with quick call option

Standard UI elements include buttons, search bars, navigation menus, and error messages for invalid inputs.

4.2 Hardware Interfaces

The application may interact with certain device hardware features:

- **GPS / Location services** to find nearby businesses
- **Phone dialer** to allow users to call businesses or emergency numbers directly
- **Mobile device storage** to store basic local settings (such as login status)

These interactions use standard mobile device APIs.

4.3 Software Interfaces

The application communicates with the following software components:

- **Supabase Backend** – used for authentication, database access, and APIs
- **PostgreSQL Database** – stores users, businesses, enquiries, offers, reviews, and other data
- **Flutter Framework** – used to build the mobile application
- **Local Storage (SharedPreferences)** – stores basic user session data on the device

The app sends and receives data such as user details, business listings, enquiries, and notifications through Supabase APIs.

4.4 Communications Interfaces

The application communicates with the backend using **secure internet connections**.

- Communication is done through **HTTPS requests** to Supabase APIs
- Data is transferred in **JSON format**
- The app requires an **active internet connection** for searching businesses, posting enquiries, and retrieving data
- Secure communication ensures safe transfer of user information.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- Business search results should load within **3 seconds** under normal mobile network conditions.
- Creating an enquiry should be completed within **5 seconds**.
- The application should support **multiple users simultaneously** without significant delays.
- The system should handle temporary network issues by retrying requests or showing clear error messages.

5.2 Safety Requirements

- The application should prevent **incorrect or incomplete data submission** by validating inputs in forms.
- Emergency numbers provided in the app should be **accurate and clearly displayed** to avoid misuse.
- The system should prevent accidental deletion of important data such as business profiles or enquiries.

5.3 Security Requirements

- Users must **authenticate using login credentials** before accessing their accounts.
- User passwords must be **stored securely using hashing**.
- **Role-based access control** should ensure customers and business owners access only their allowed features.
- Secure communication using **HTTPS** must be used for all backend API requests.
- Database access should be controlled using **Row Level Security (RLS)** policies.

5.4 Software Quality Attributes

Usability: The application should be easy to use with a simple and intuitive mobile interface.

Reliability: The system should handle errors gracefully and maintain stable operation during normal usage.

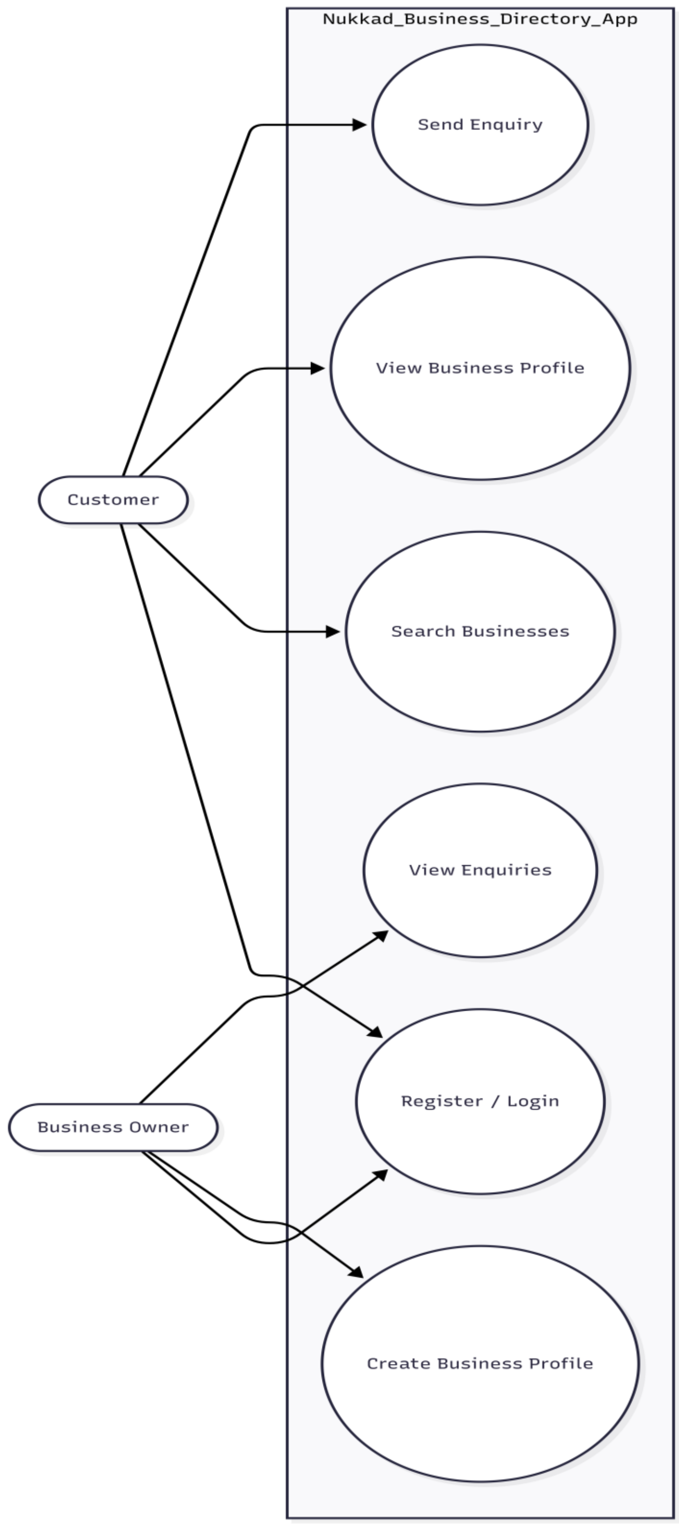
Maintainability: The application should follow a modular structure where services handle backend communication, making future updates easier.

Portability: The application should run on **Android, iOS, and web platforms** using Flutter.

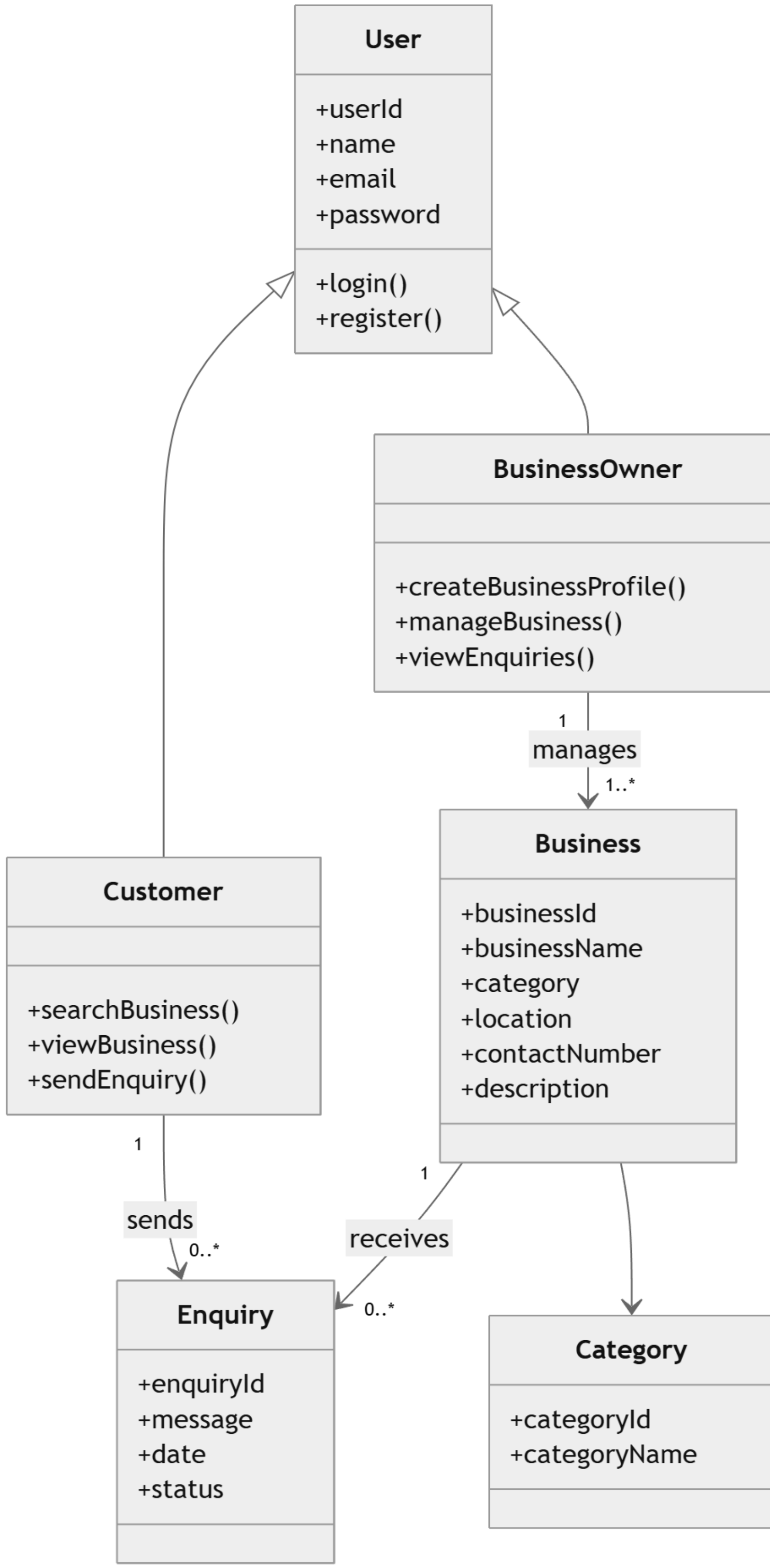
Availability: The system should remain accessible whenever internet connectivity is available.

6. Diagrams

6.1 Use Case Diagram

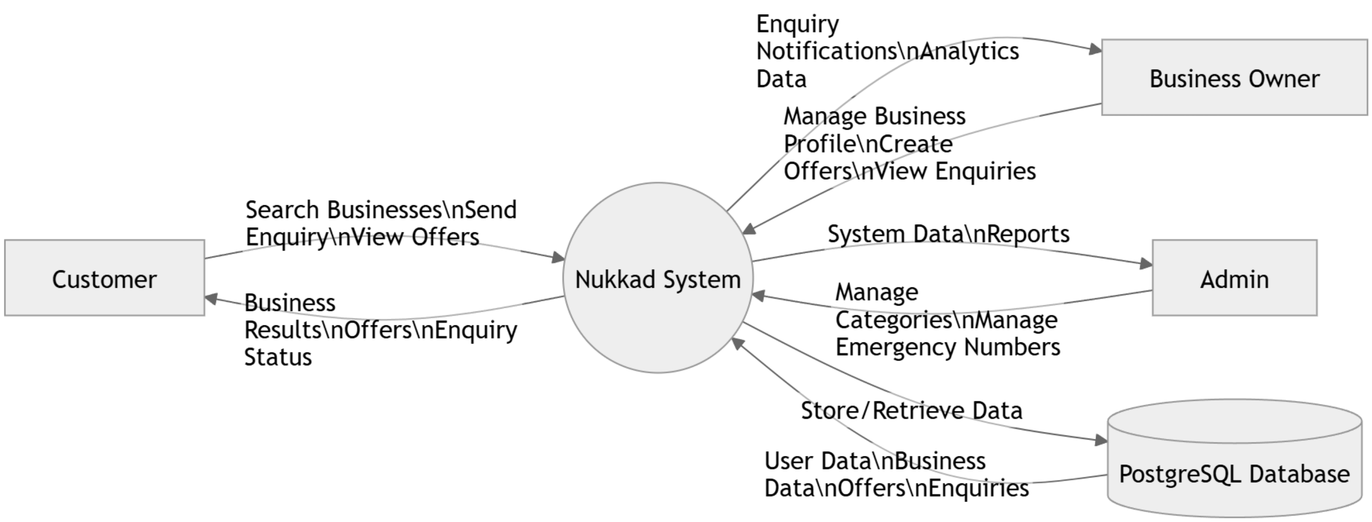


6.2 Class Diagram

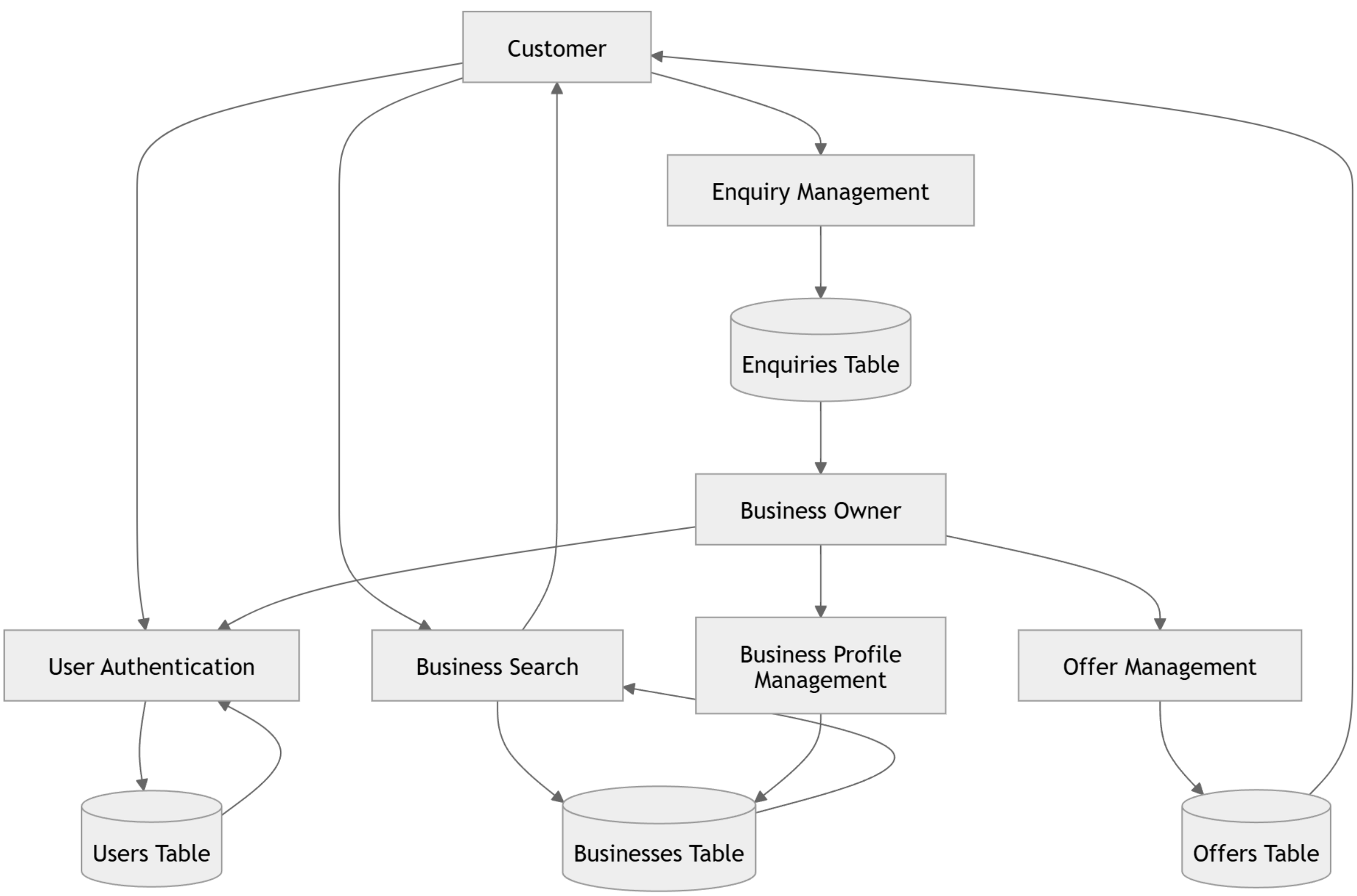


6.3 Data Flow Diagram

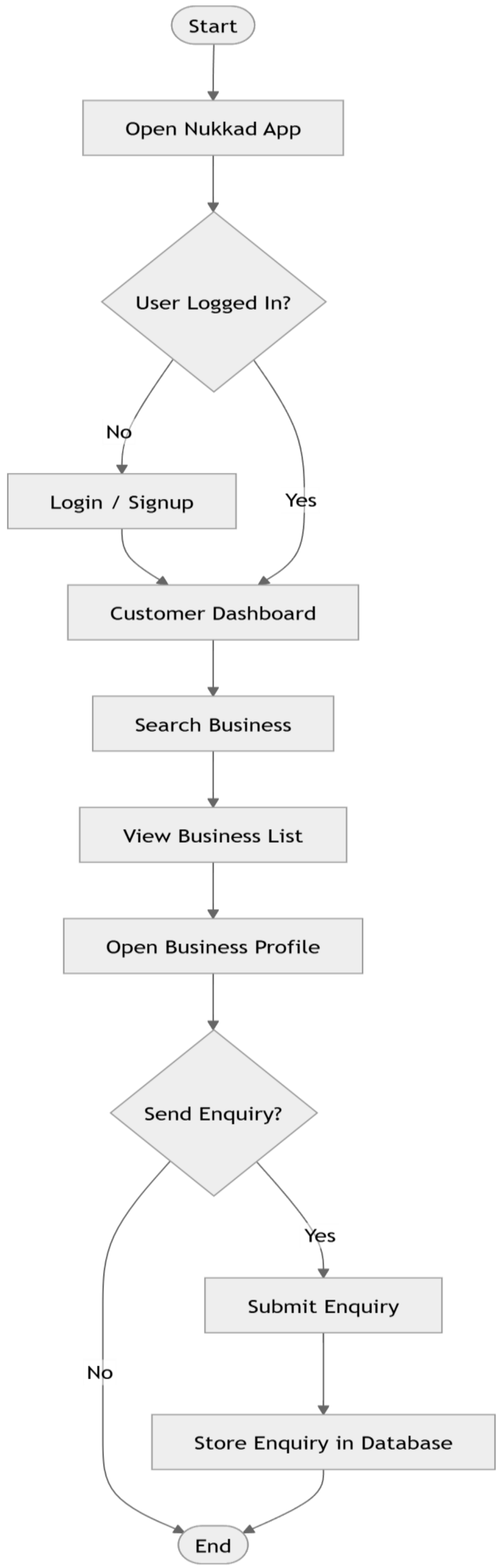
DFD Level 0 – Context Diagram:



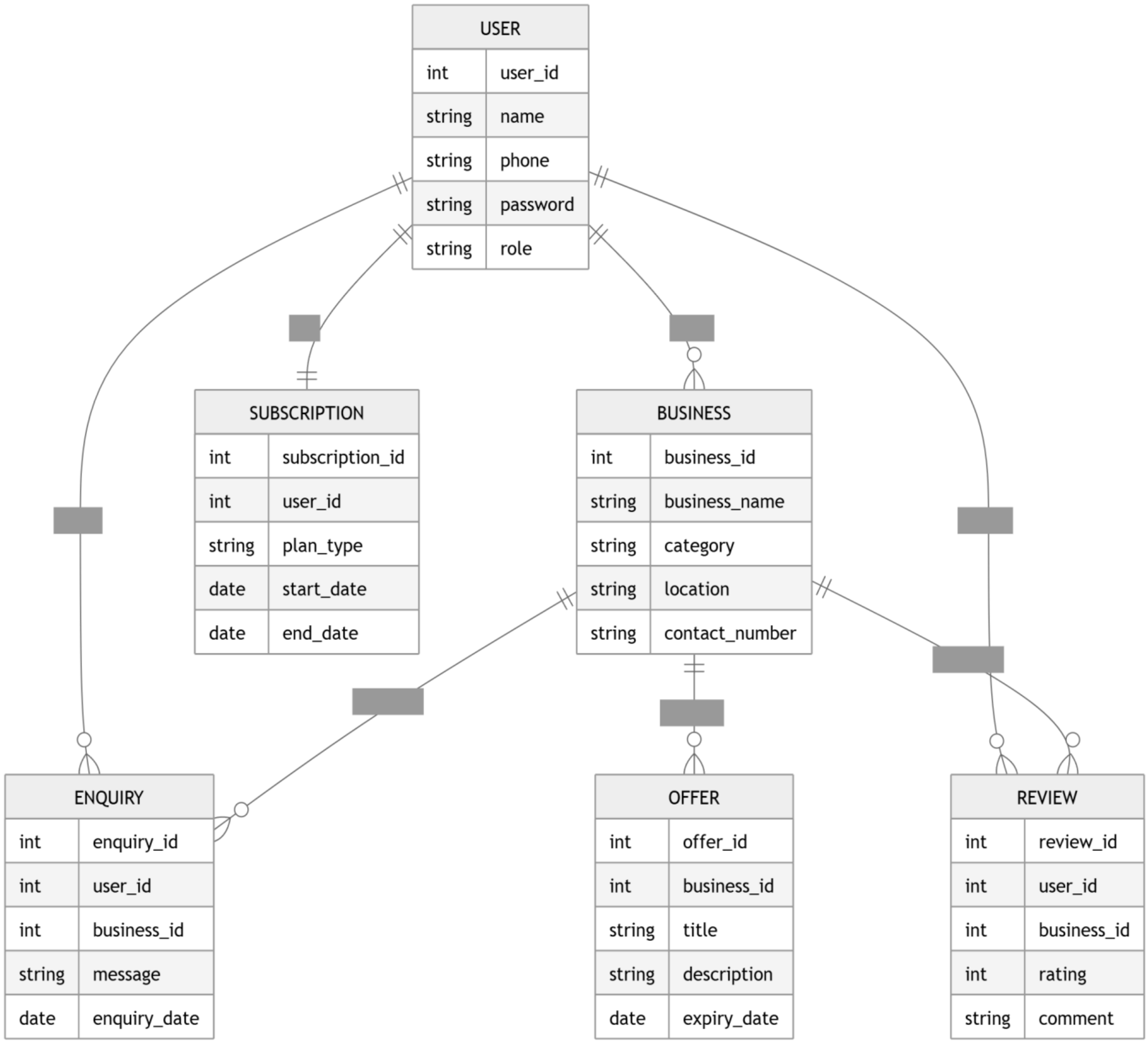
DFD Level 1:



6.4 Activity Diagram



6.5 ER Diagram



7. Other Requirements

Database Requirements

- The system uses a PostgreSQL database through Supabase.
- Data such as users, businesses, enquiries, offers, subscriptions, and reviews must be stored in structured tables.
- Relationships between tables must be maintained using foreign keys.
- Database triggers are used to automatically update connects when enquiries are created.

Legal Requirements

- The application must ensure user data privacy and protection.
- Users should only access their own data based on their roles.

Internationalization

- The application will initially support English language.
- Future versions may support multiple languages.

Appendix A: Glossary

App – The Nukkad Business Directory mobile application.
Business Owner – A user who registers and manages their business profile in the app.
Customer – A user who searches for businesses and sends enquiries.
Connect – A credit used by a customer when posting an enquiry to a business.
Supabase – Backend platform used for database, authentication, and APIs.
Enquiry – A request or message sent by a customer to a business.
Offer – A promotion or deal created by a business owner.

Appendix B: Analysis Models

The following models are used to describe the system design:

- **Use Case Diagram** – Shows interactions between customers, business owners, and the system.
- **Data Flow Diagram (DFD) –** Shows how data moves between users, processes, and the database.
- **Entity Relationship Diagram (ERD)** – Shows relationships between database tables such as users, businesses, enquiries, and subscriptions.

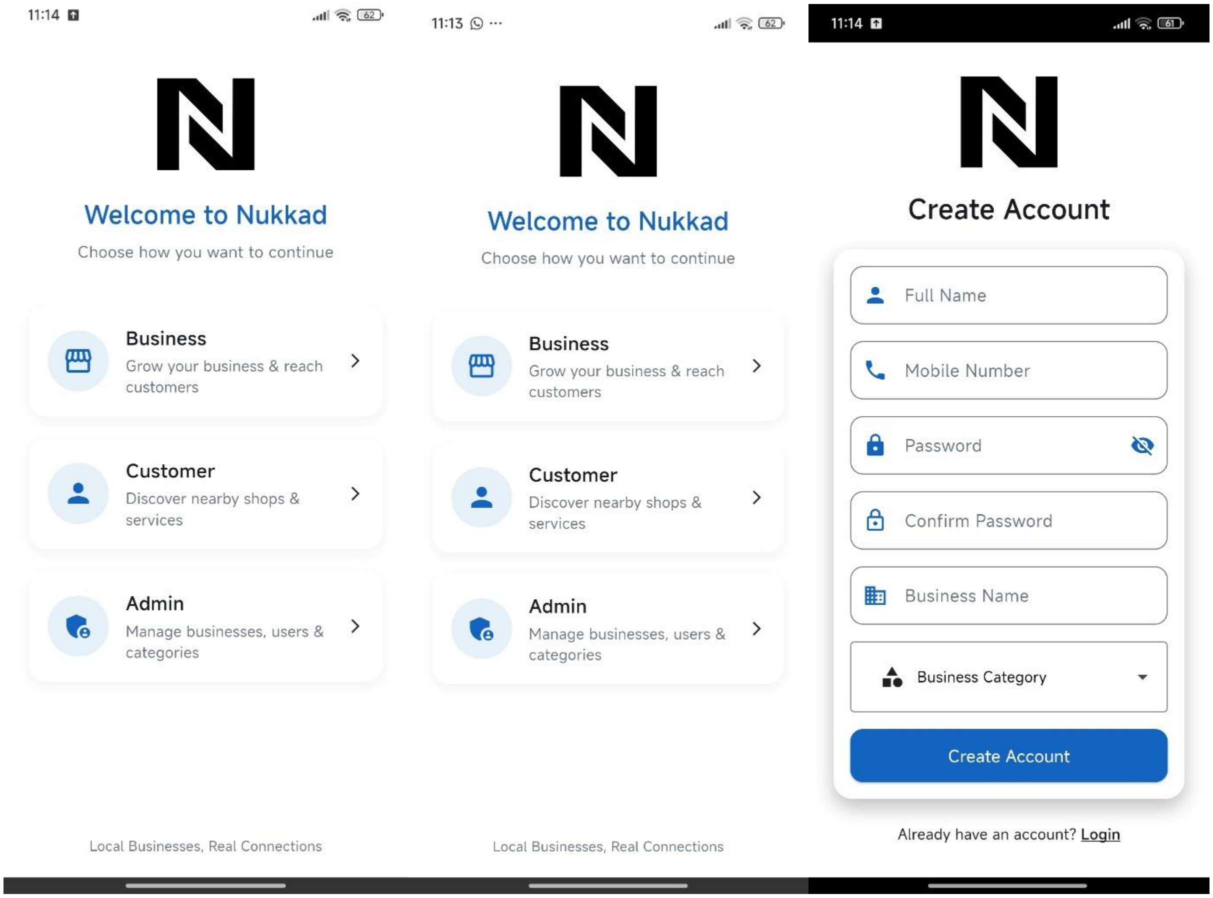
Appendix C: Issues List

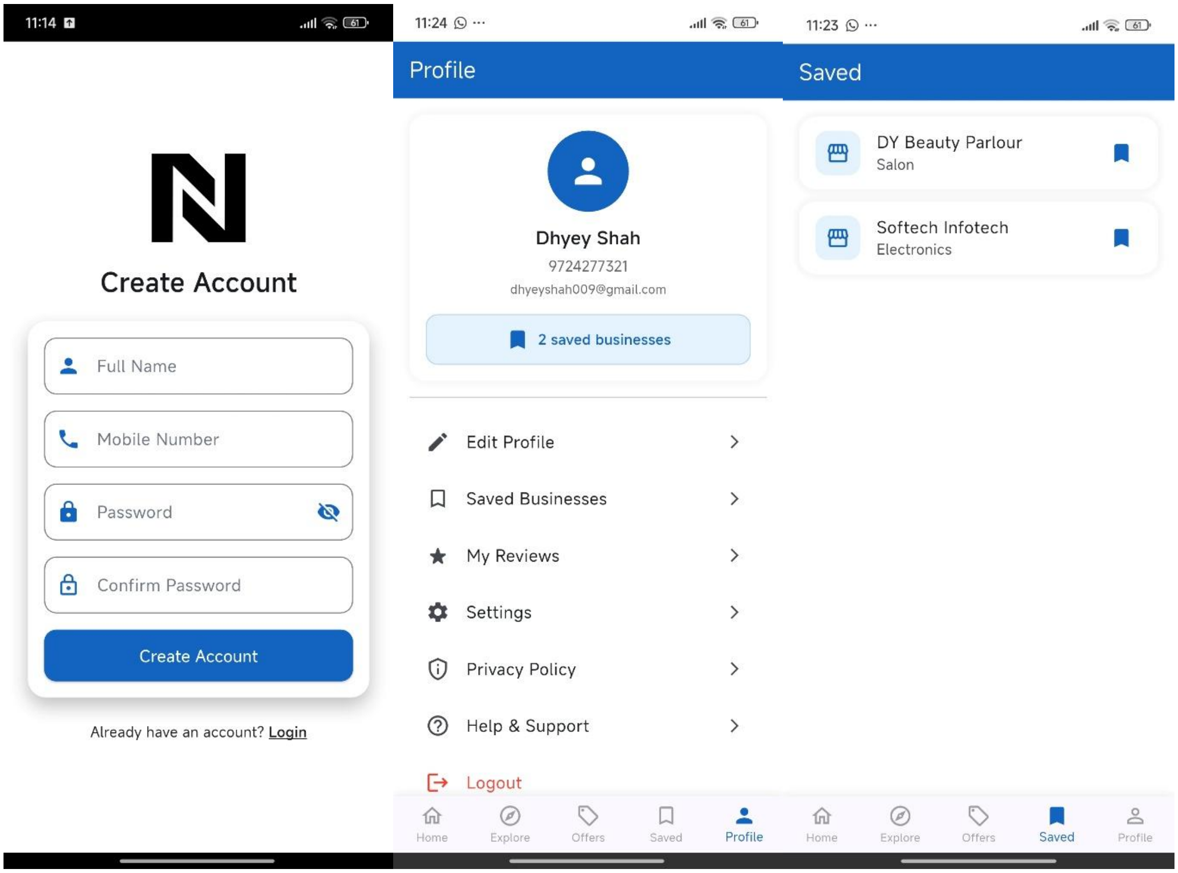
- **Payment gateway integration** for subscriptions is not yet finalized.
- **Admin dashboard UI** is not implemented in the current version.
- Some notification features are **still to be defined (TBD)**.

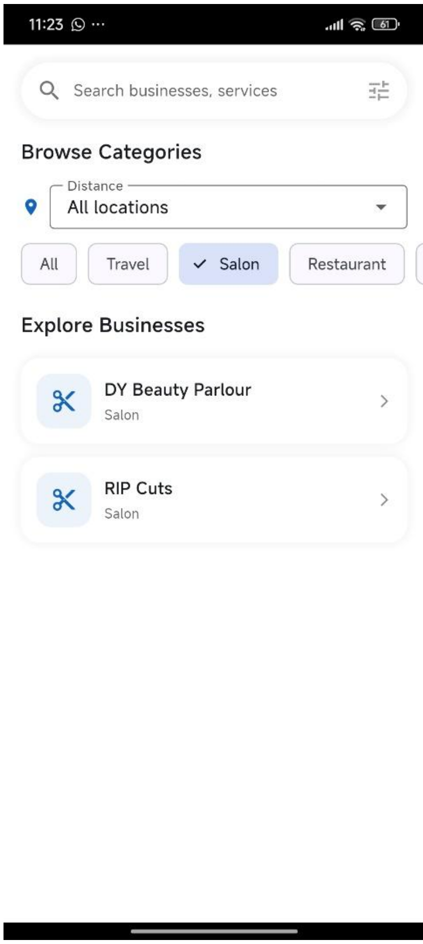
Future Improvements

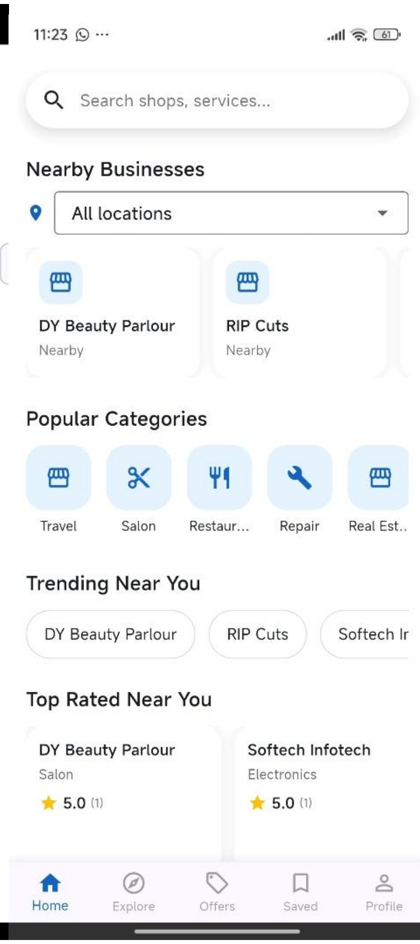
The Nukkad Business Directory can be further improved by integrating advanced features such as AI-based recommendation systems to suggest businesses based on user preferences and past activity. Personalized suggestions can enhance user experience and engagement. Additionally, improved analytics dashboards for business owners can provide insights into customer behavior, enquiry trends, and offer performance. Future versions may also include smart search, location-based recommendations, and enhanced notification systems to make the platform more efficient and intelligent.

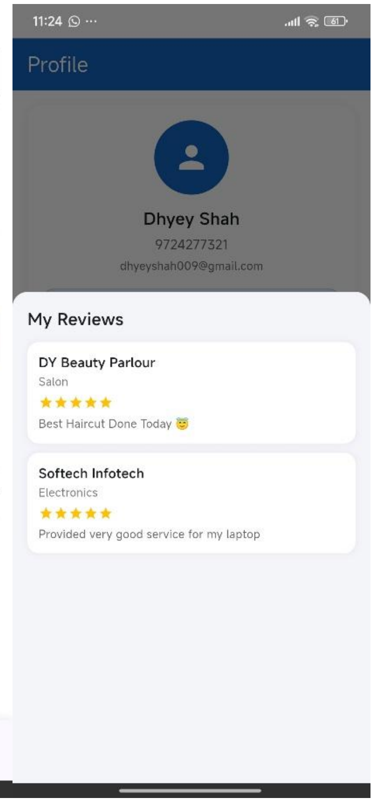
8. User Interface

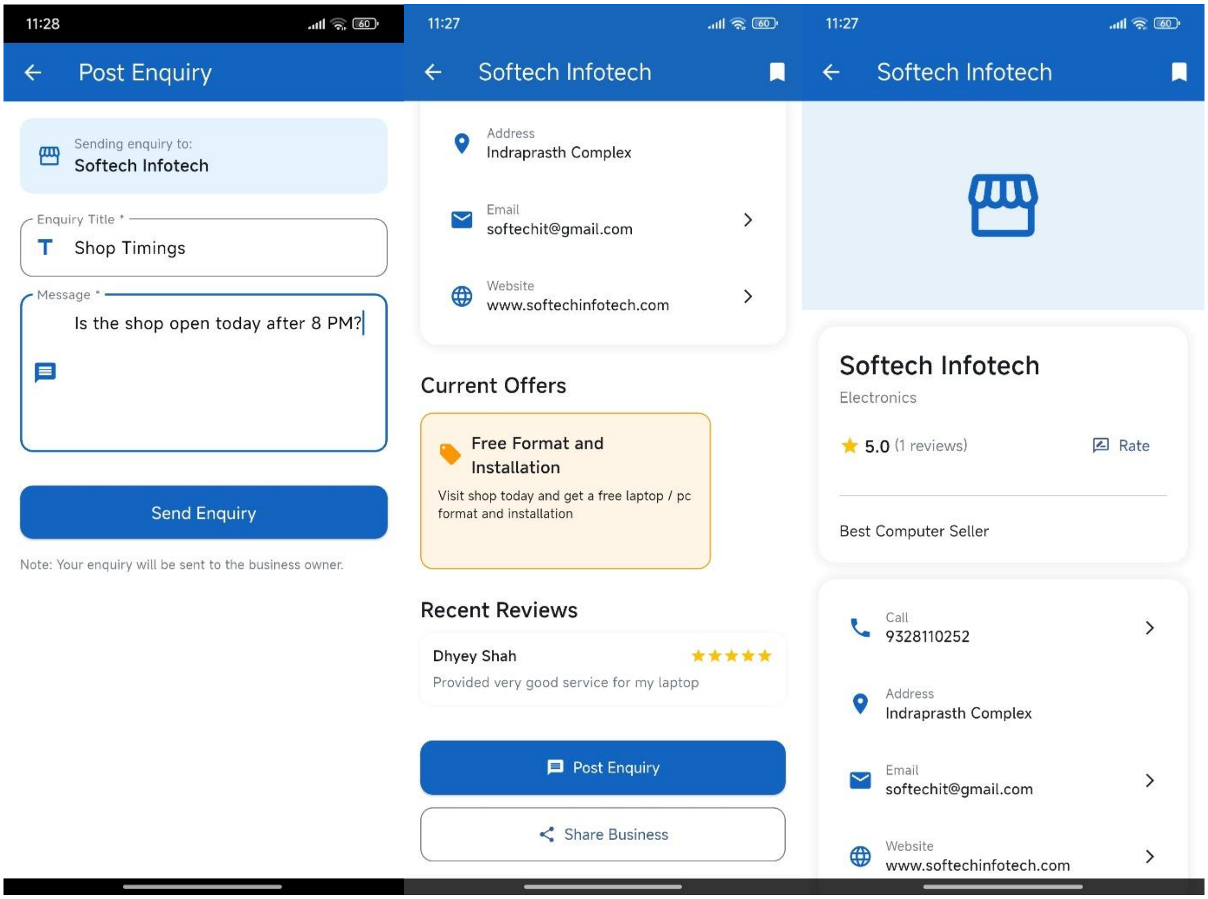


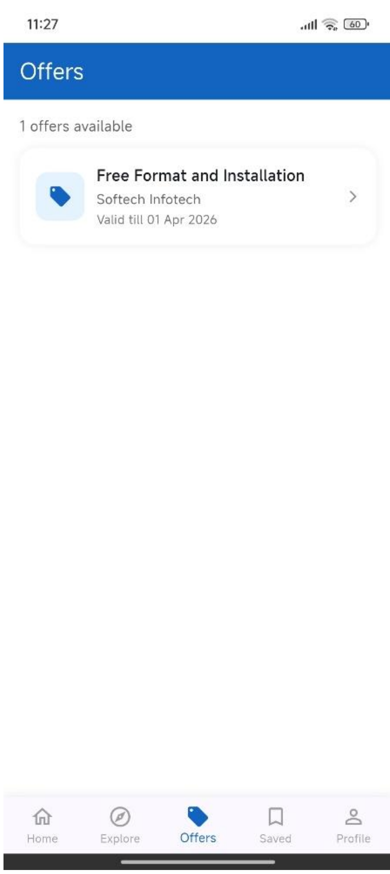

11:14
Create Account
Full Name
Mobile Number
Password
Confirm Password
Create Account
Already have an account? Login
11:24
Profile
Dhyey Shah
9724277321
dhyeyshah009@gmail.com
2 saved businesses
Edit Profile
Saved Businesses
My Reviews
Settings
Privacy Policy
Help & Support
Logout
Home
Explore
Offers
Saved
Profile
11:23
Saved
DY Beauty Parlour
Salon
Softech Infotech
Electronics
Home
Explore
Offers
Saved
Profile


11:23
Search businesses, services
Browse Categories
Distance
All locations
All
Travel
Salon
Restaurant
Explore Businesses
DY Beauty Parlour
Salon
RIP Cuts
Salon


11:23
Search shops, services...
Nearby Businesses
All locations
DY Beauty Parlour
Nearby
RIP Cuts
Nearby
Popular Categories
Travel
Salon
Restaur...
Repair
Real Est..
Trending Near You
DY Beauty Parlour
RIP Cuts
Softech Ir
Top Rated Near You
DY Beauty Parlour
Salon
5.0 (1)
Softech Infotech
Electronics
5.0 (1)
Home
Explore
Offers
Saved
Profile


11:24
Profile
Dhyey Shah
9724277321
dhyeyshah009@gmail.com
My Reviews
DY Beauty Parlour
Salon
Best Haircut Done Today
Softech Infotech
Electronics
Provided very good service for my laptop


11:28
Post Enquiry
Sending enquiry to:
Softech Infotech
Enquiry Title *
Shop Timings
Message *
Is the shop open today after 8 PM?
Send Enquiry
Note: Your enquiry will be sent to the business owner.
11:27
Softech Infotech
Address
Indraprasth Complex
Email
softechit@gmail.com
Website
www.softechinfotech.com
Current Offers
Free Format and Installation
Visit shop today and get a free laptop / pc format and installation
Recent Reviews
Dhyey Shah
Provided very good service for my laptop
Post Enquiry
Share Business
11:27
Softech Infotech
Softech Infotech
Electronics
5.0 (1 reviews)
Rate
Best Computer Seller
Call
9328110252
Address
Indraprasth Complex
Email
softechit@gmail.com
Website
www.softechinfotech.com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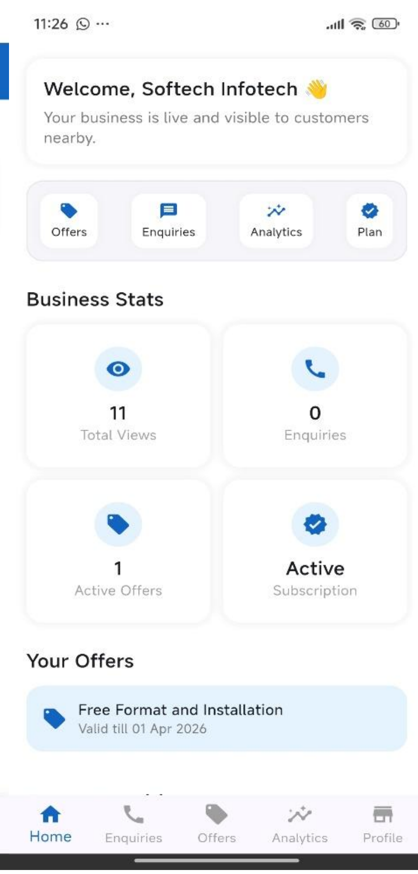
Offers
1 offers available
Free Format and Installation
Softech Infotech
Valid till 01 Apr 2026
Home
Explore
Offers
Saved
Profile


11:26
Welcome, Softech Infotech 👋
Your business is live and visible to customers nearby.
Offers
Enquiries
Analytics
Plan
Business Stats
11
Total Views
0
Enquiries
1
Active Offers
Active
Subscription
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Free Format and Installation
Valid till 01 Apr 2026
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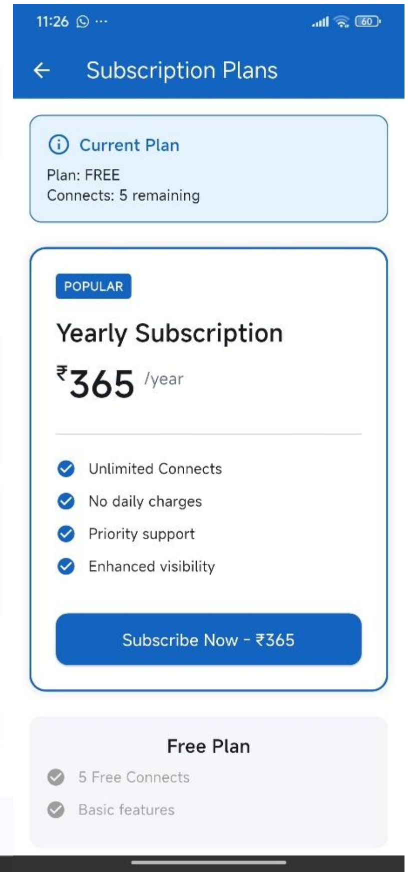
Enquiries
Offers
Analytics
Profile


11:26
Subscription Plans
Current Plan
Plan: FREE
Connects: 5 remaining
POPULAR
Yearly Subscription
₹365 /year
Unlimited Connects
No daily charges
Priority support
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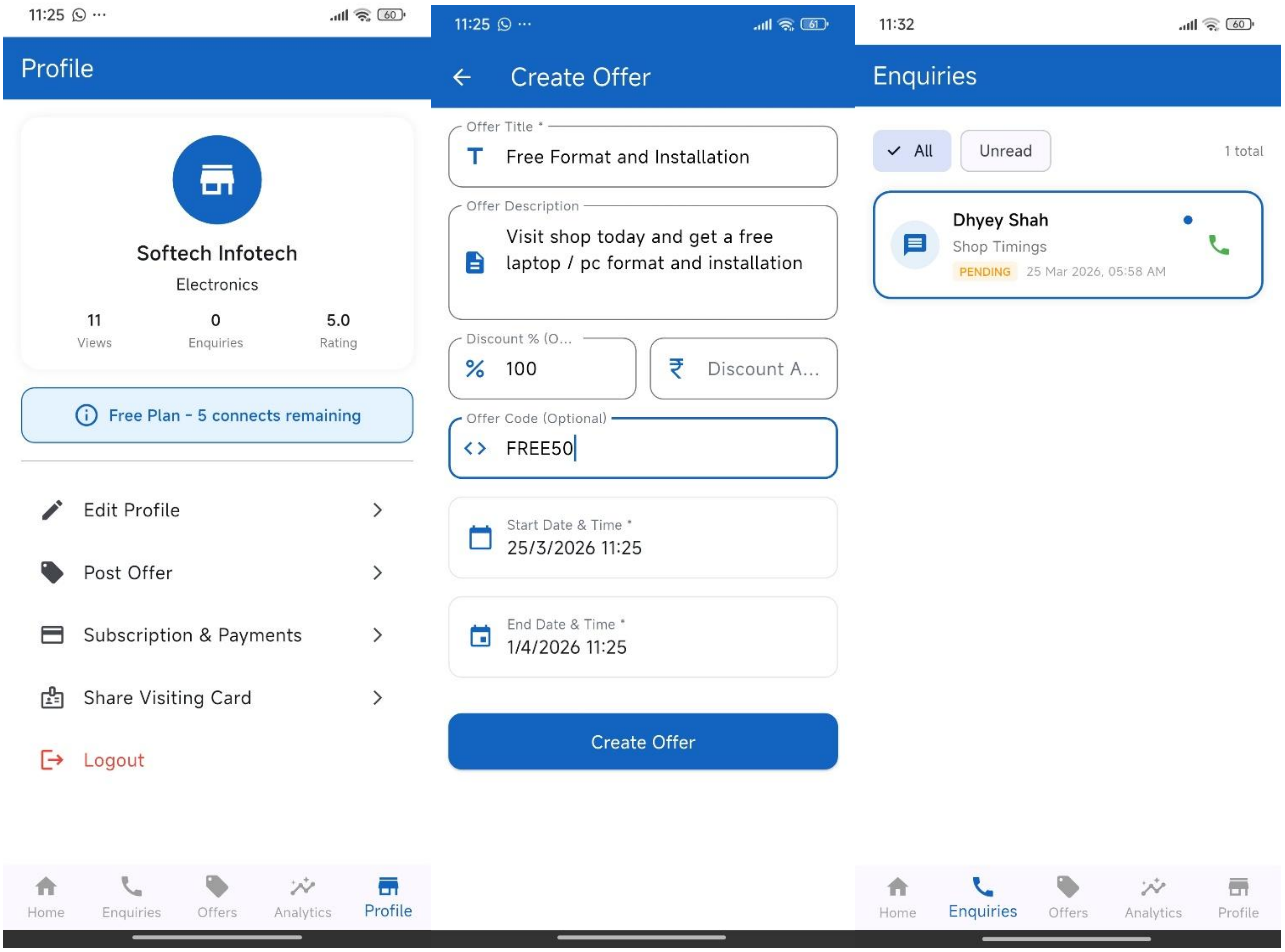
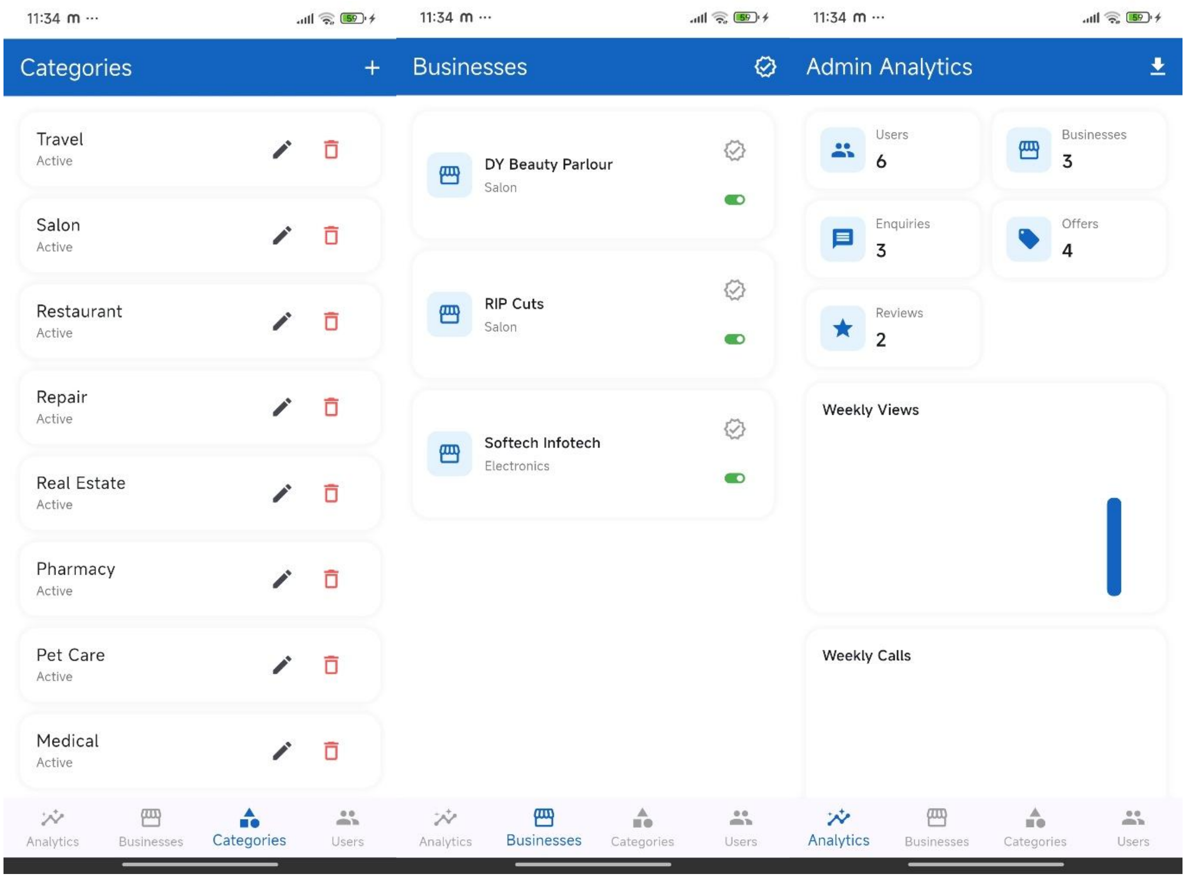
Free Plan
5 Free Connects
Basic features


11:25
Profile
Softech Infotech
Electronics
11
Views
0
Enquiries
5.0
Rating
Free Plan - 5 connects remaining
Edit Profile
Post Offer
Subscription & Payments
Share Visiting Card
Logout
Home
Enquiries
Offers
Analytics
Profile
11:25
Create Offer
Offer Title *
Free Format and Installation
Offer Description
Visit shop today and get a free laptop / pc format and installation
Discount % (O...
100
Discount A...
Offer Code (Optional)
FREE50
Start Date & Time *
25/3/2026 11:25
End Date & Time *
1/4/2026 11:25
Create Offer
11:32
Enquiries
All
Unread
1 total
Dhyey Shah
Shop Timings
PENDING
25 Mar 2026, 05:58 AM
Home
Enquiries
Offers
Analytics
Profile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Repair
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Real Estate
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Pharmacy
Active
Pet Care
Active
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Active
Analytics
Businesses
Categories
Users
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Businesses
DY Beauty Parlour
Salon
RIP Cuts
Salon
Softech Infotech
Electronics
Analytics
Businesses
Categories
Users
11:34
Admin Analytics
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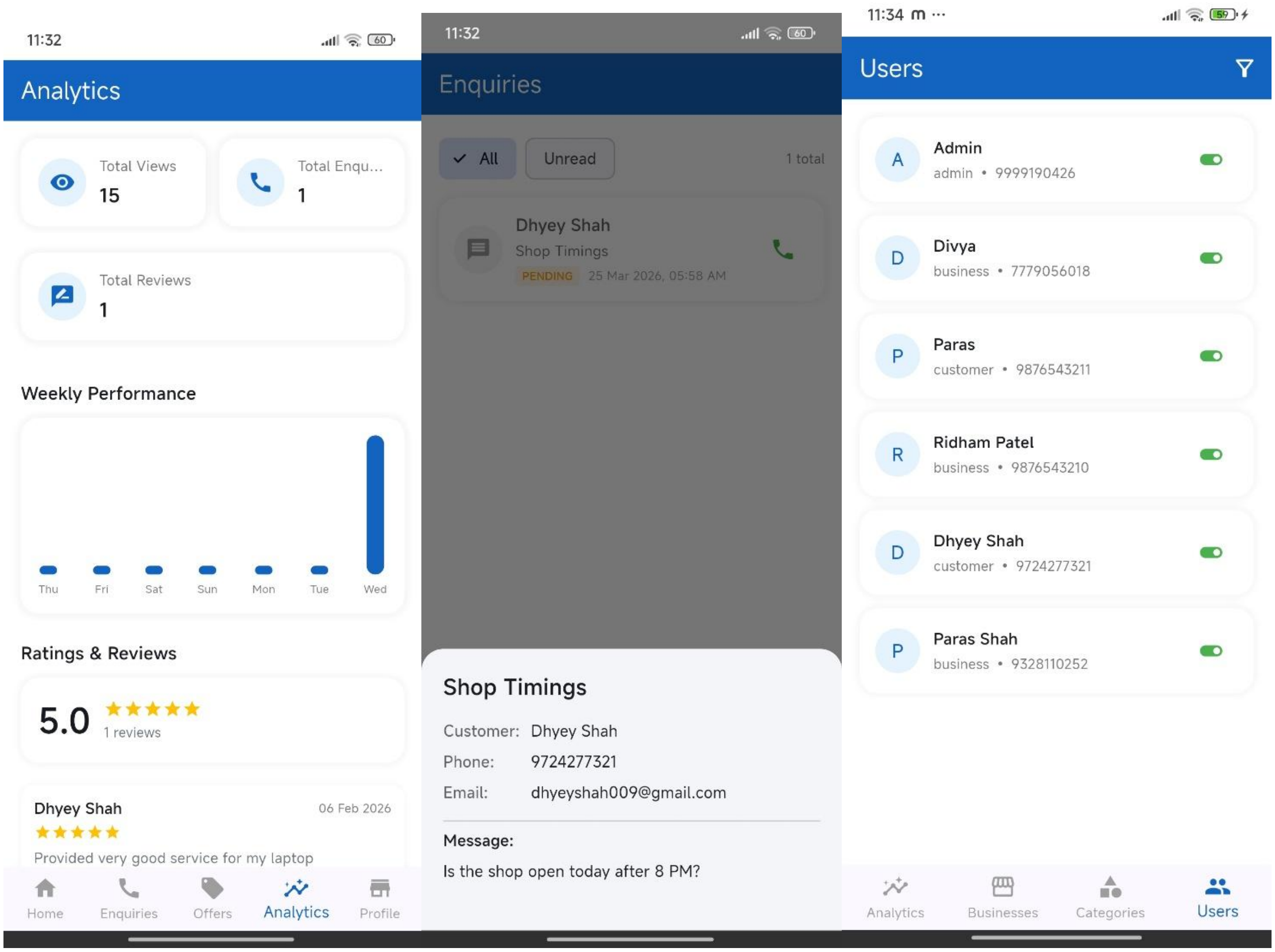
Analytics
Businesses
Categories
Users



Conclusion:

The **Nukkad** Business Directory project successfully demonstrates the design and development of a system that connects customers with local businesses efficiently. Through this project, key concepts of system design, database management, and mobile application development were applied. It also provided practical experience in building scalable features like authentication, search, and enquiry management. Overall, the project enhanced understanding of real-world application development and problem-solving in a structured manner.